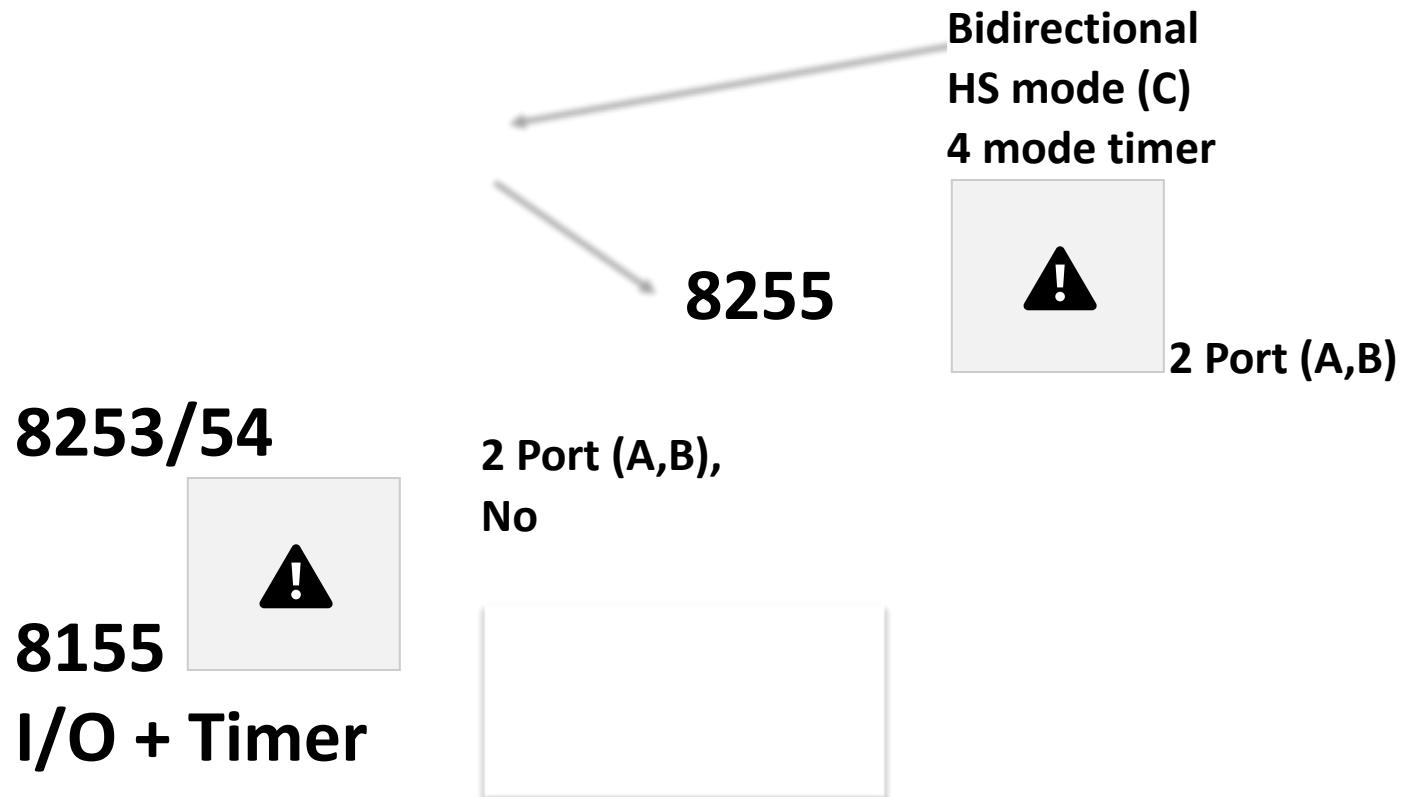


Programming 8251 USART Controller

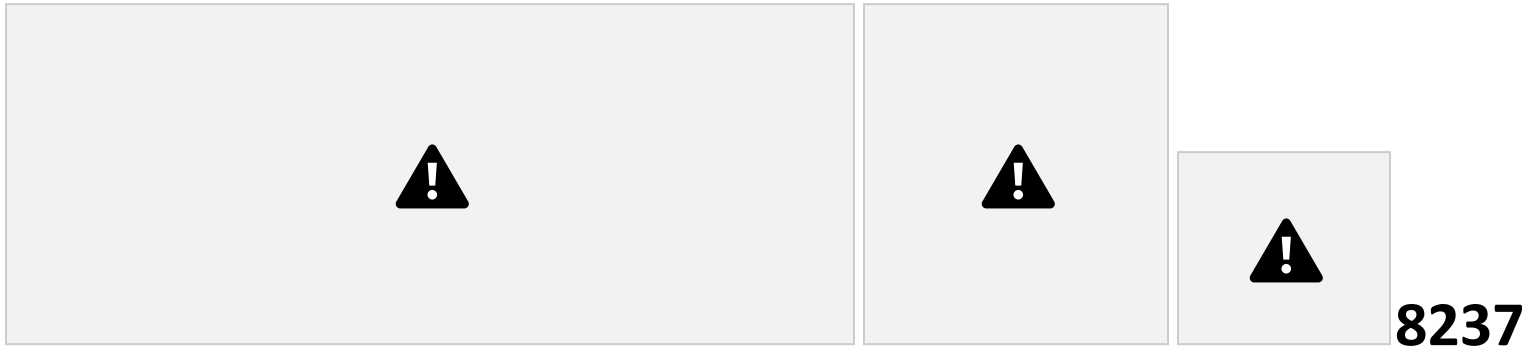
Hierarchy of I/O Control Devices



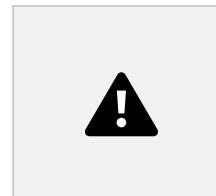


Timer 8259

6 mode timer



I/O



A is Bidirectional HS mode (C)

Extra controls **8251**



DMA controller

**Serial I/O USART
controller**

Interrupt controller



- Asynchronous Communication
- 8251 USART Architecture

- USART Registers
- Programming UART
- RS 232 Port
- Interfacing CRT Monitor using a UART and RS 232 port



- Serial
 - Faster
 - Cheaper
 - Slower
 - Data skew
- Parallel



**Data
Transmission**

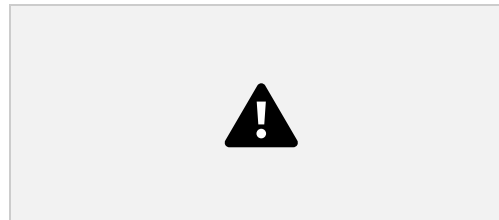
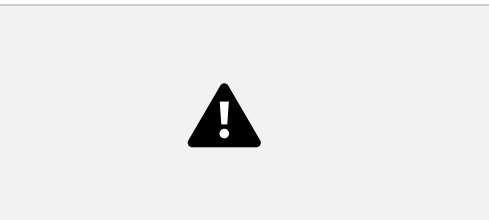


Serial

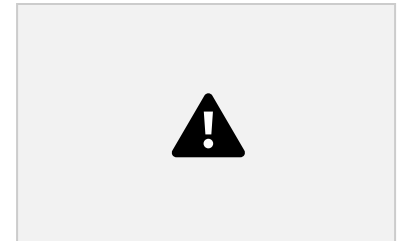


Parallel

– Limited to small



distances



Synchronous
Asynchronous

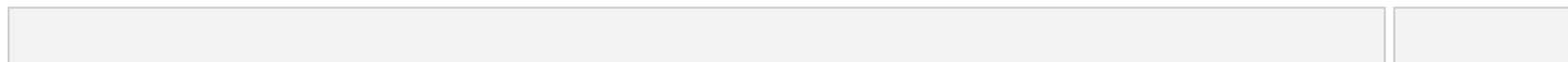


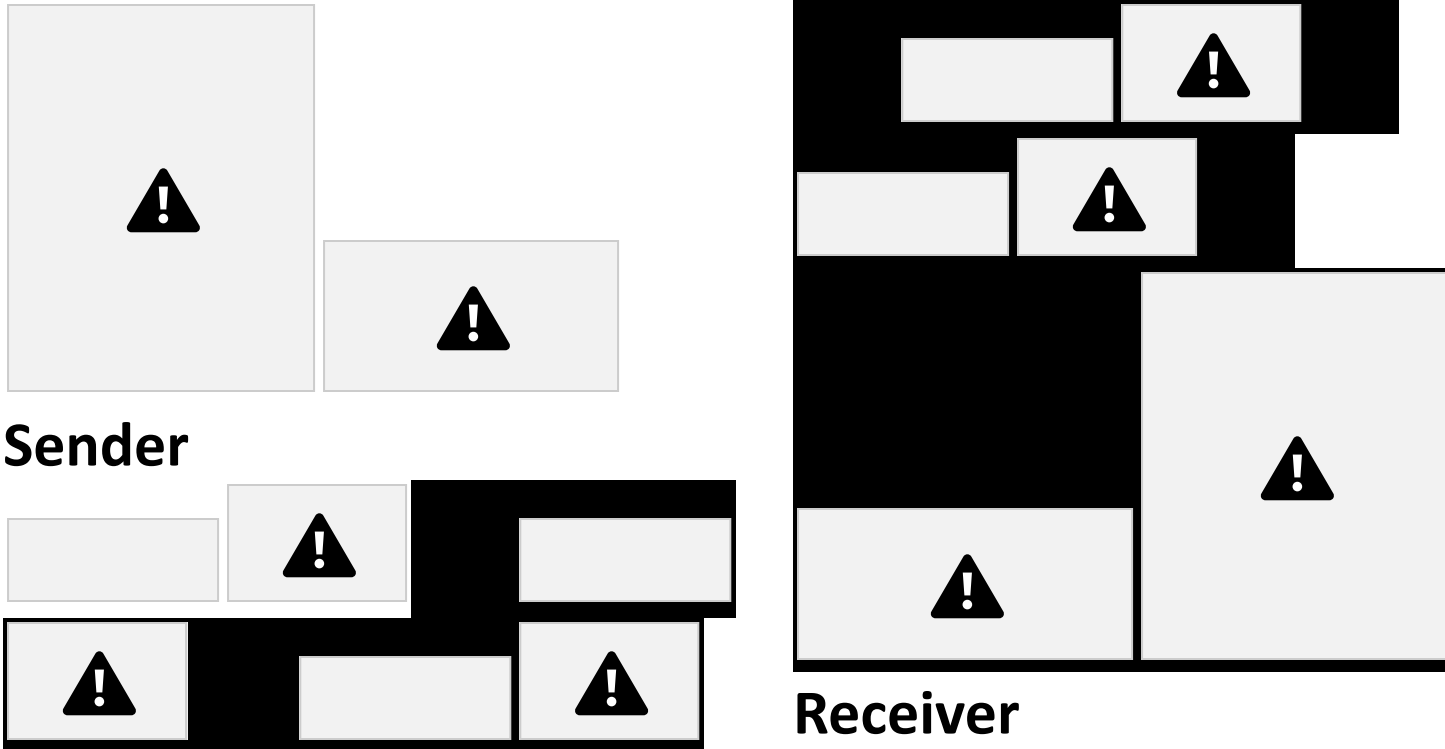
Transmission Gaps
Sender

Receiver

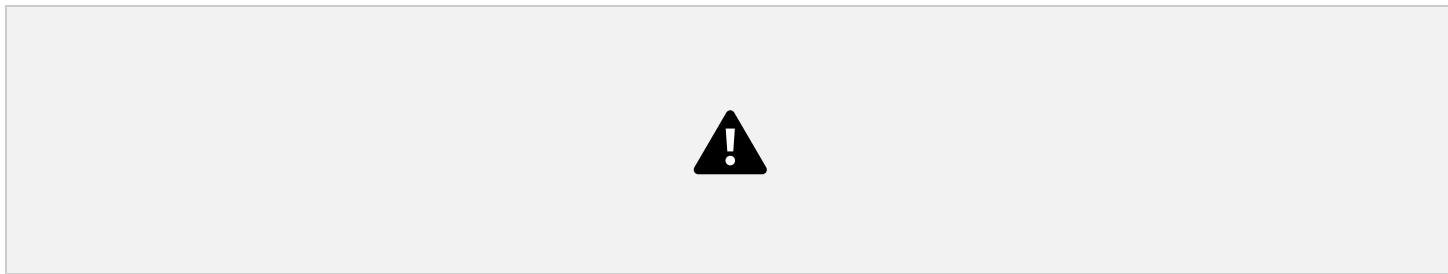
Asynchronous transmission

CLK





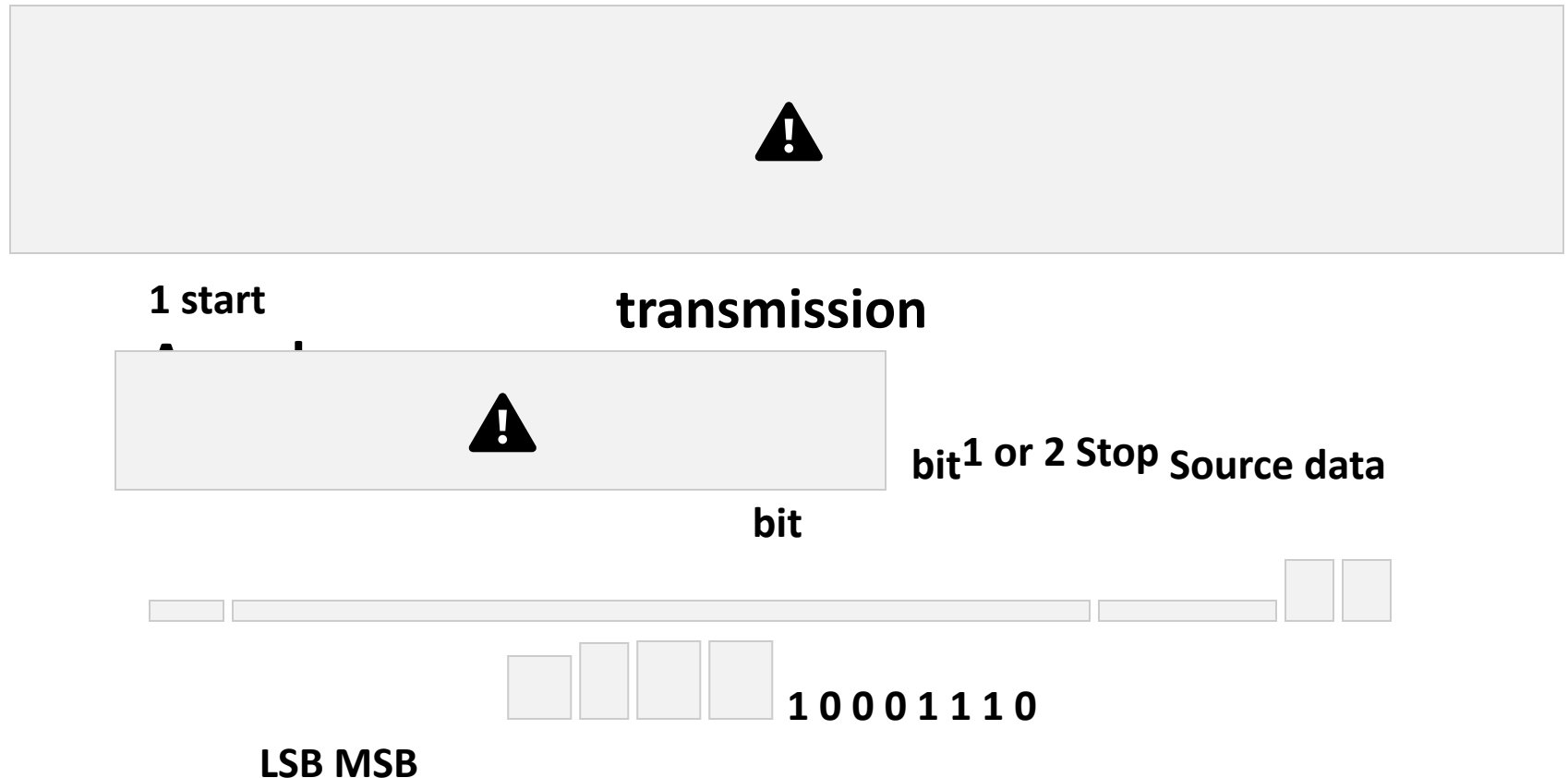
Synchronous transmission



- Character oriented
- Each character carried start bit and stop bits •

When No data are being transmitted – Receiver stay at logic 1 called mark, logic 0 is Space • Framing:

- Transmission begins with one start bit (low/0)
- Followed by DATA (8bit) and
- Stop bits (1 or 2 bits of logic high)





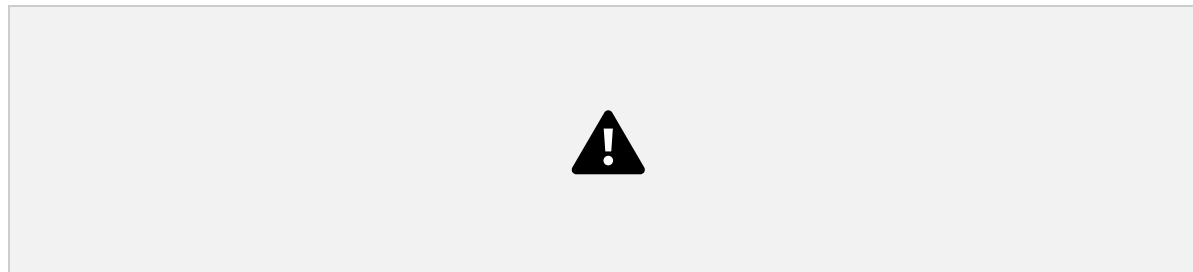
Time

Start Bit Start Bits



8

bit Data



- Serial Input Data (SID)

- Serial Output Data (SOD)

- Instruction SIM is necessary to output data
- Interpretations (ACC contents)

SOD	SDE (0/1 Dis/Ena SOD)	X	For interrupts
-----	--------------------------	---	-------------------



MVI A, 80 ; Set D₇ in the
ACC=1

RAR ; Set D₆ =1
and bring carry

into D₇ SIM ; output D₇





d in Register B

MVI B ASCIIDataByte ;
get data byte in B
MVI C,0BH ; set up
counter for 11 bits
XRA A ; reset carry to 0
NXTbit: MVI A,80H ;set
D7=1 in ACC
RAR
;bring
Carry in
D7 and
set
D6=1
SIM
;output

D7

CALL DELAYBittime ;wait for fixed time (BWT)

STC ;set Carry 1

MOV A,B ;Place ASCII car in acc

RAR ; place ASCII D0 in Carry

;and shift 1 in D7

```
MOV B,A ;Save B
DCR C
JNZ NXTbit
RET
```



- Programmable chip 8251
- Requirement of HW control serial I/O – An input/output port are required for interfacing – Converts data bits in to Parallel to serial & vice versa
 - Data transfer to be synchronized between I/O
 - USART (Universal Synchronous Asynchronous Receiver and Transmitter)

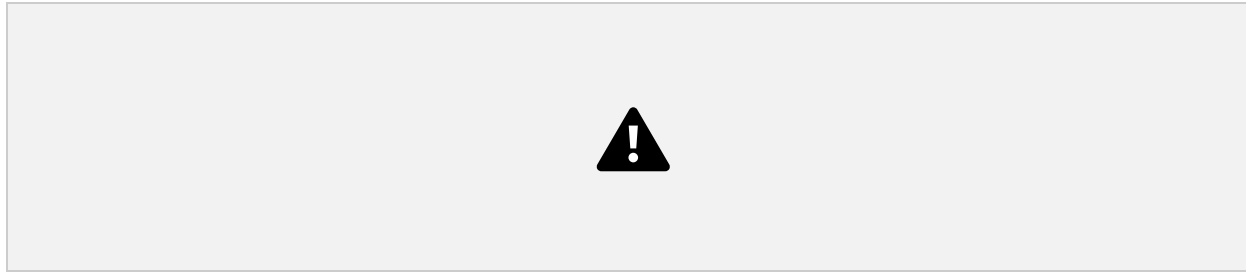
- Writing a program compatible with all different serial



communication protocols is difficult and it is an inefficient use of microprocessor.

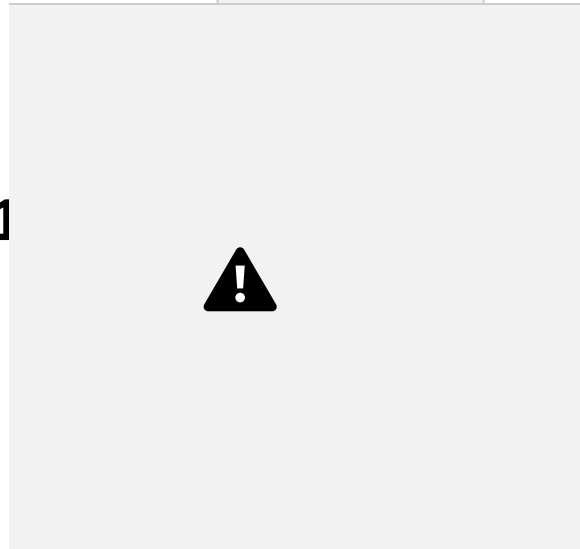
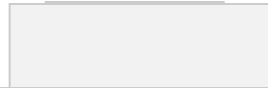
- UART: Universal Asynchronous Receiver/Transmitter chip.
- USART: Universal Synchronous/Asynchronous Receiver/Transmitter chip.
- The microprocessor sends/receives the data to the UART in parallel, while with I/O, the UART transmits/receive data serially.
- 8251 functions are integrated into standard PC

interface chip.



CPU

8251



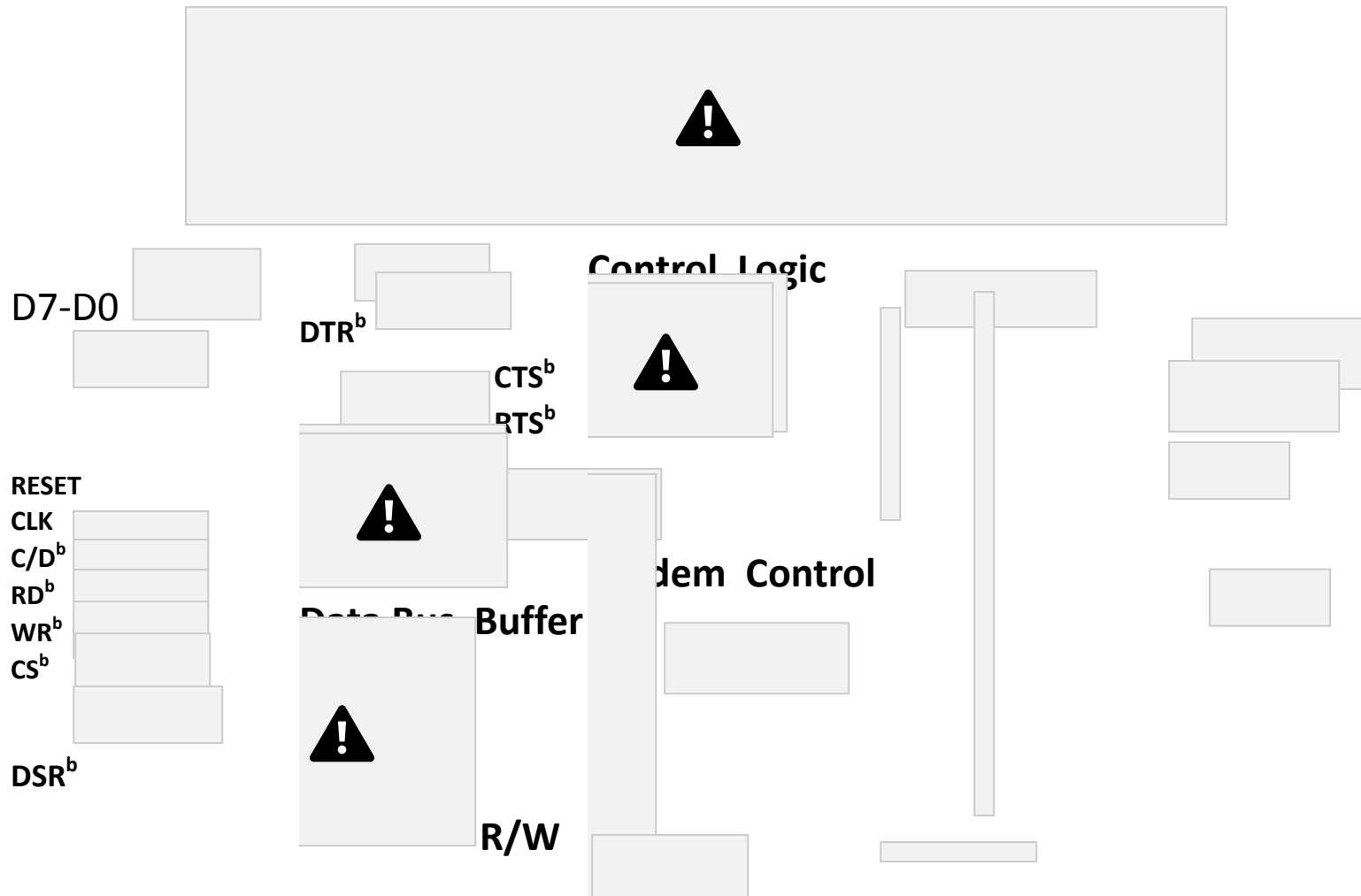
**(8 bit)
serial port**

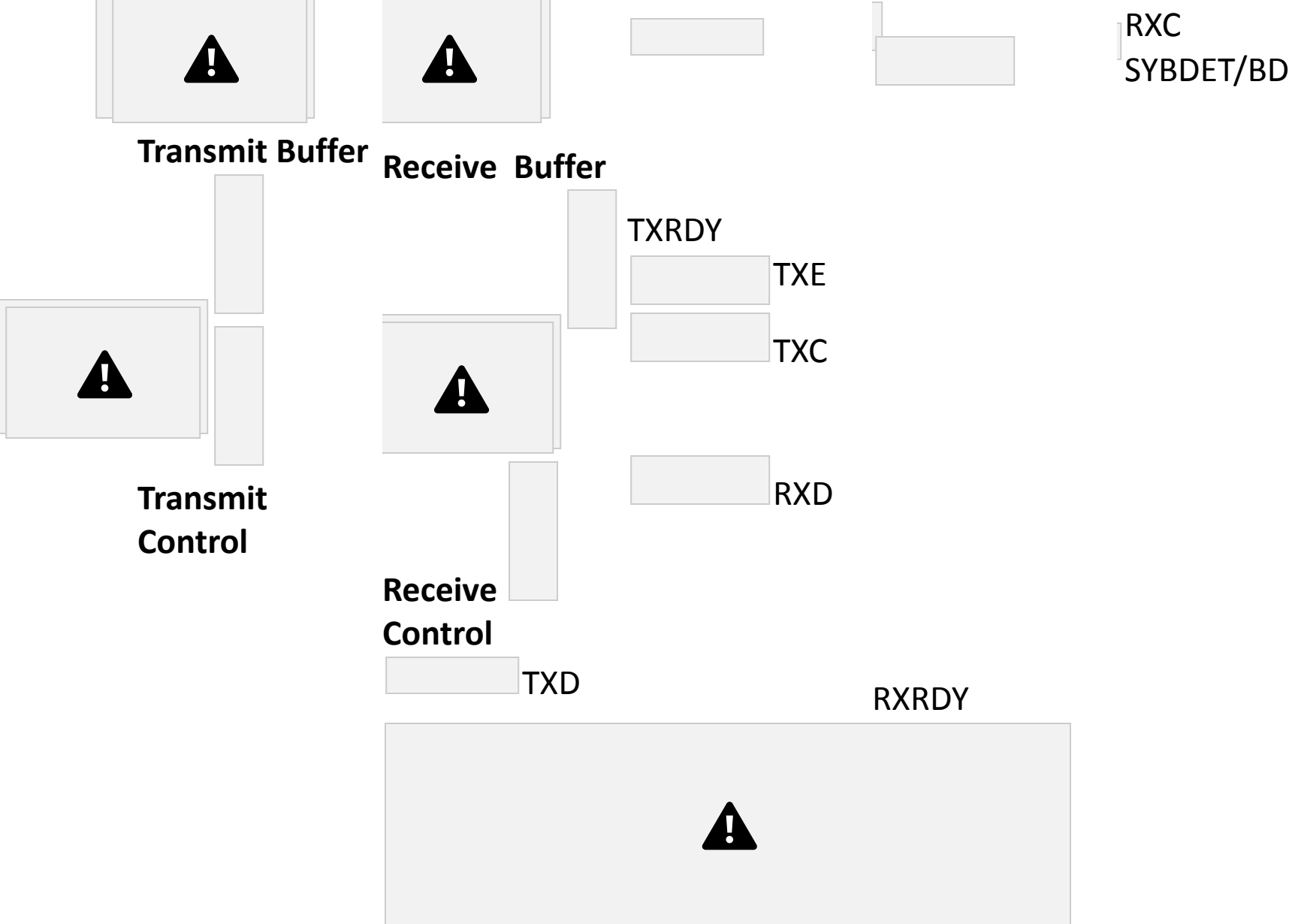
- UART/USART •
- 8251 USART

data



- 8250/16450 UART is a newer version of 8251.
- 16550 is the latest version UART.





CS	C/D	RD	WR	Meaning
1	X	X	X	<i>Data Bus Tri-state</i>
0	X	1	1	Data Bus Tri-state
0	1	0	1	Status □ CPU
0	1	1	0	Control Word □ CPU
0	0	0	1	Data □ CPU (accept data from Data Buffer)
0	0	1	0	Data □ CPU (Out put data to Data buffer)



RD^b or WR^bControl

D7-D0

C/D^b=0

Data Buffer

register

Transmitter

RESET CLK

C/D^b

RD^b

WR^b

CS^b

R/W

Control Logic

C/D^b=1

WR^b=0

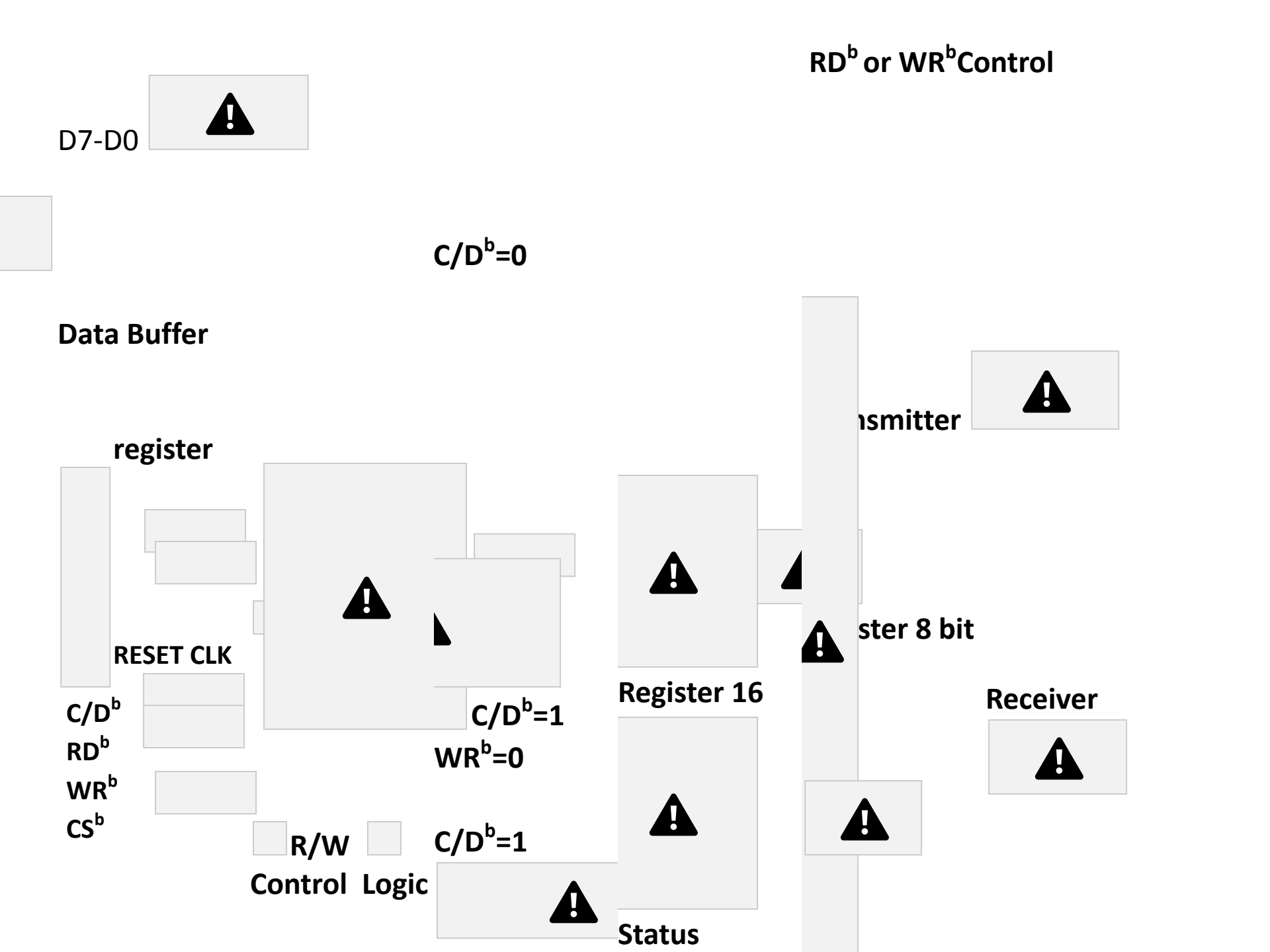
C/D^b=1

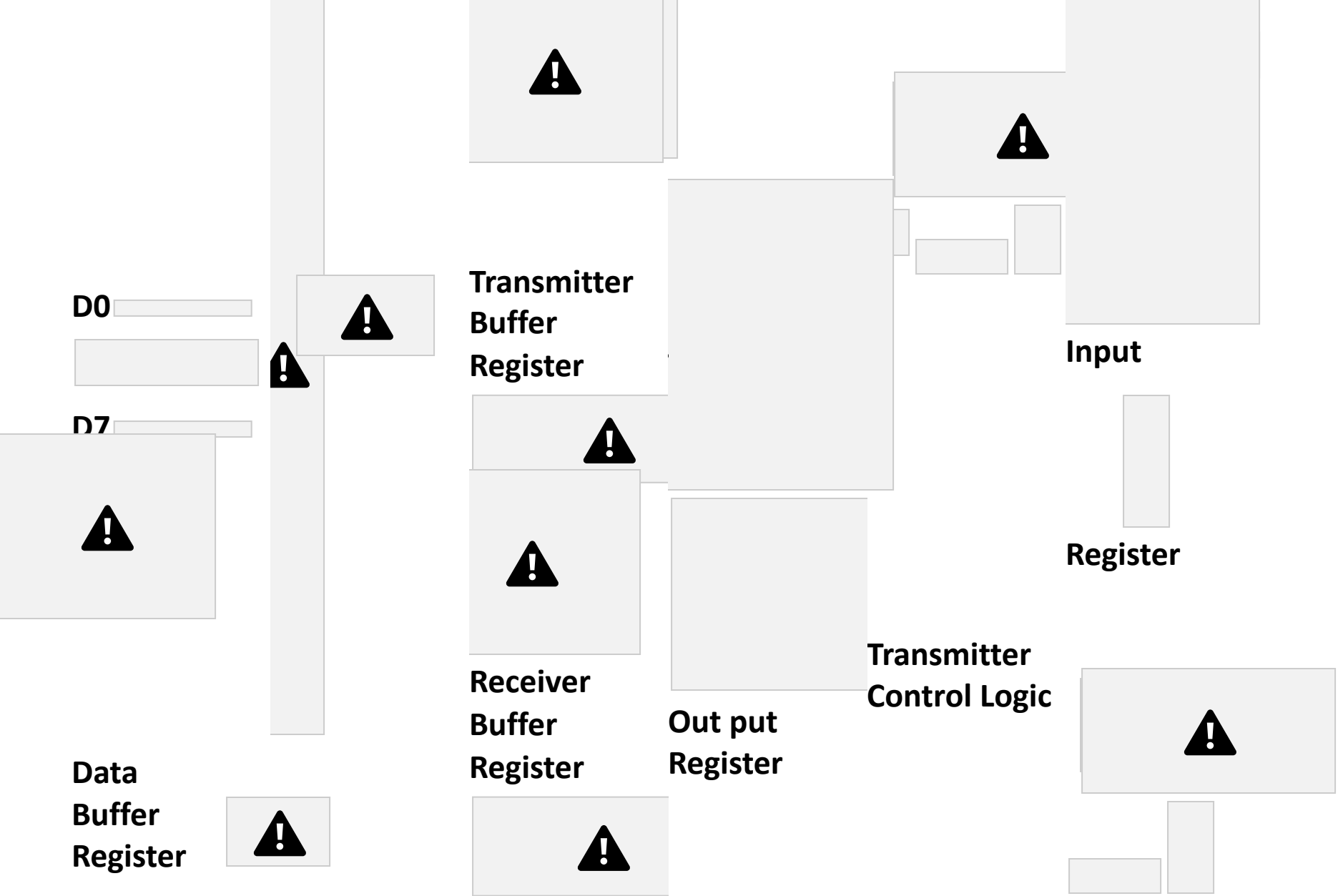
Register 16

Register 8 bit

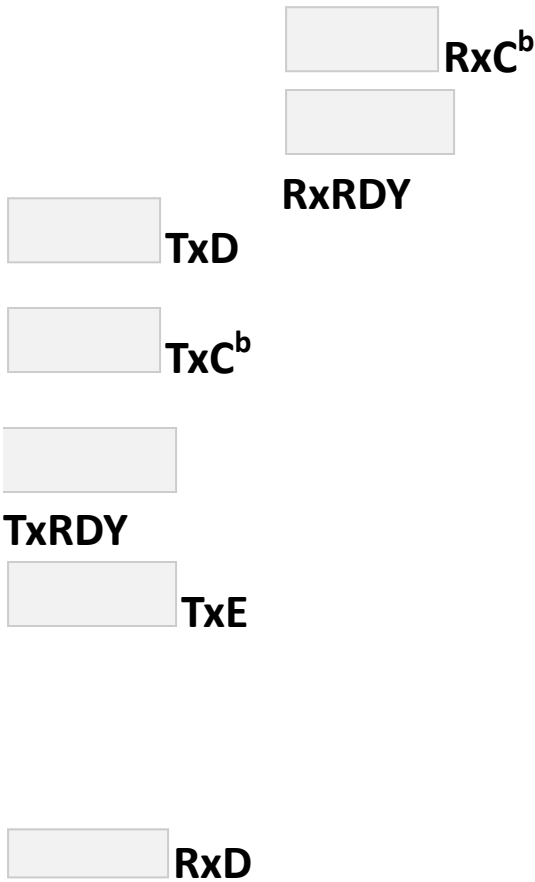
Receiver

Status

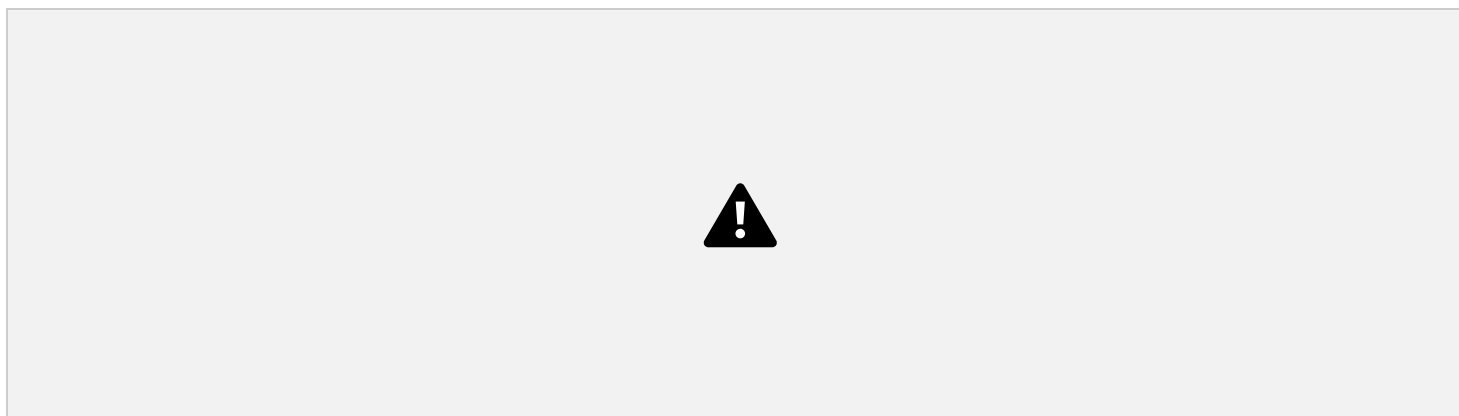
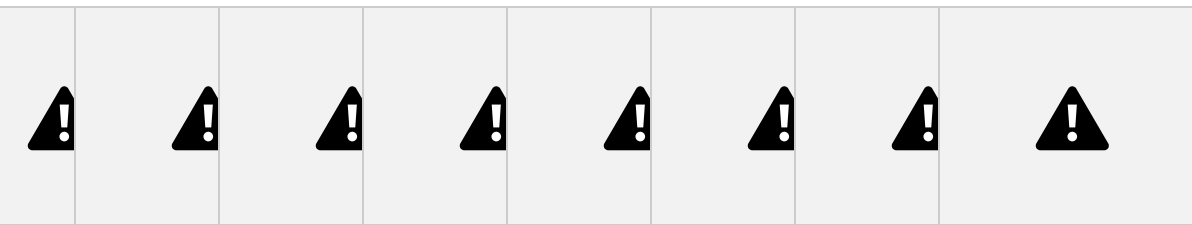




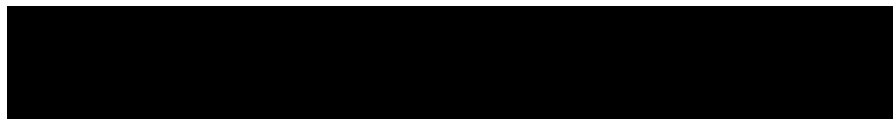
Control Logic



Receiver



D7





Framing Control

of Stop bits

00: invalid

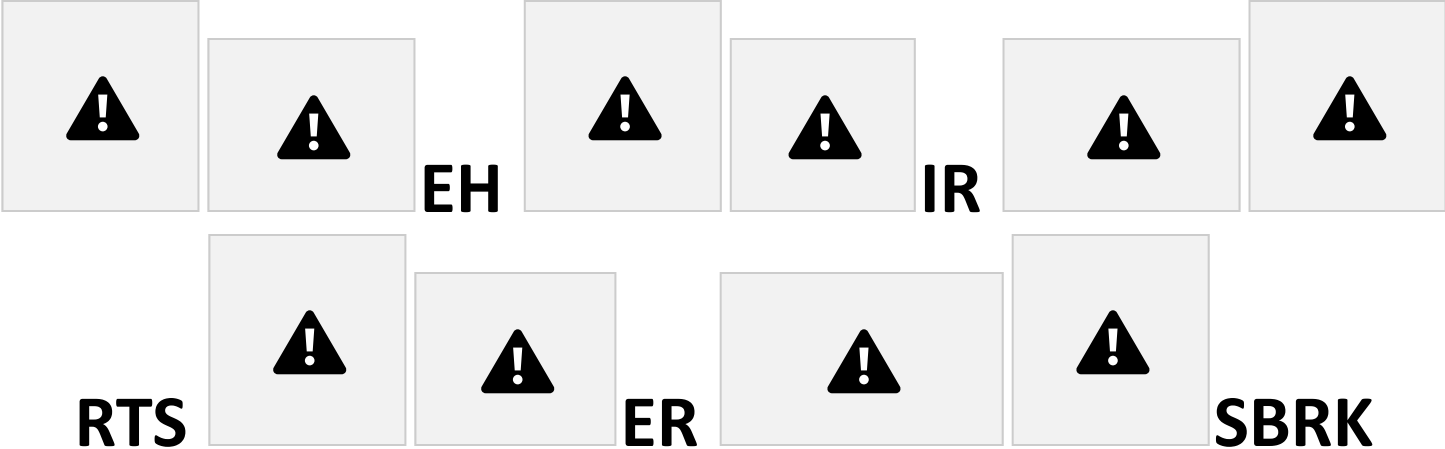
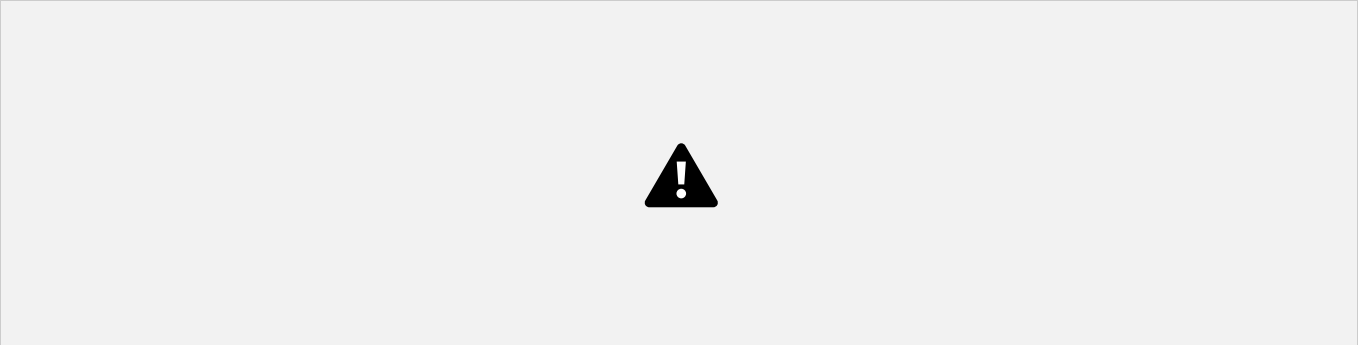
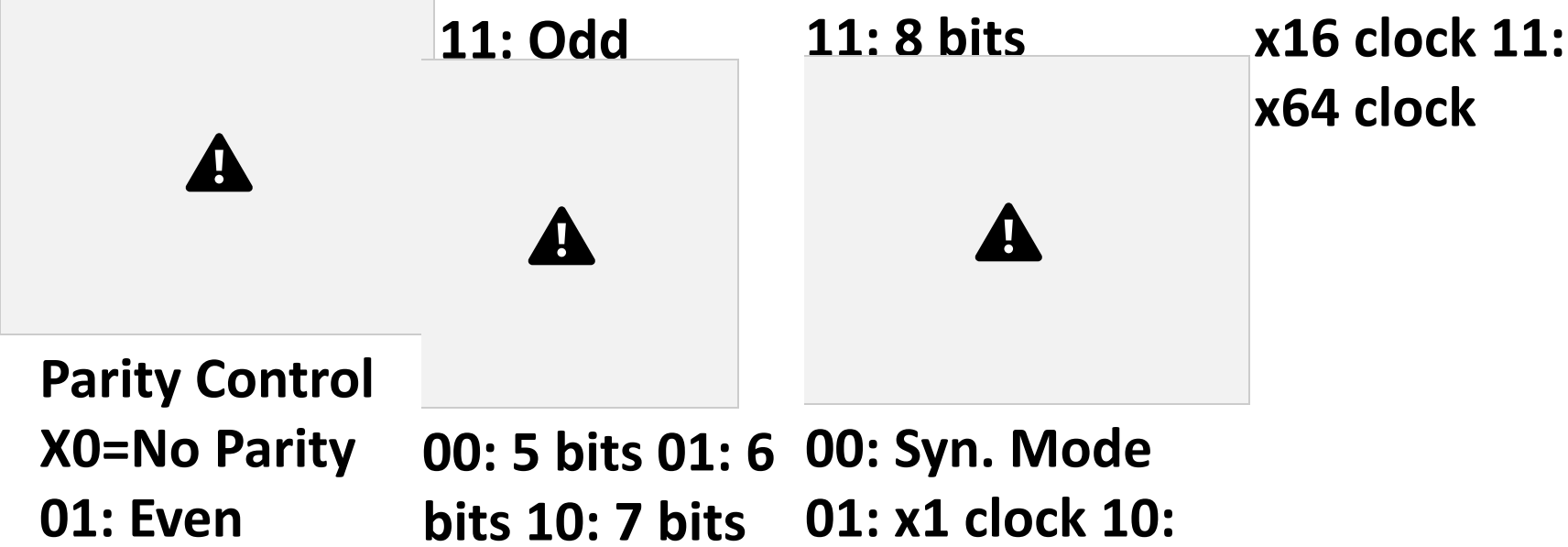
Baud

Character length

Rate

bits 11: 2 bits

01: 1 bit 10: 1.5





TxE TxE: transmit enable (0/1 Enable Disable) DTR: data terminal ready (1=ENABLE DTR) Rx E: receiver enable (1/0=EN/DISABLE) SBPRK: send break character 1= force TxD low ER: error reset (Reset Flags: Parity ,Over run, Framing Error of Status Word) RTS: request to send (1= Enable Request to send) IR: internal reset (Reset 8251 to mode) EH: enter hunt mode (1=search for Sync Character)



DSR SYN

DET FE OE PE Tx

EMPTY RxRDY TxRDY

TxRDY transmit ready (DB Buffer is empty)

RxRDY receiver ready

TxEMPTY transmitter empty

PE parity error (1=when PE detected) OE

overrun error

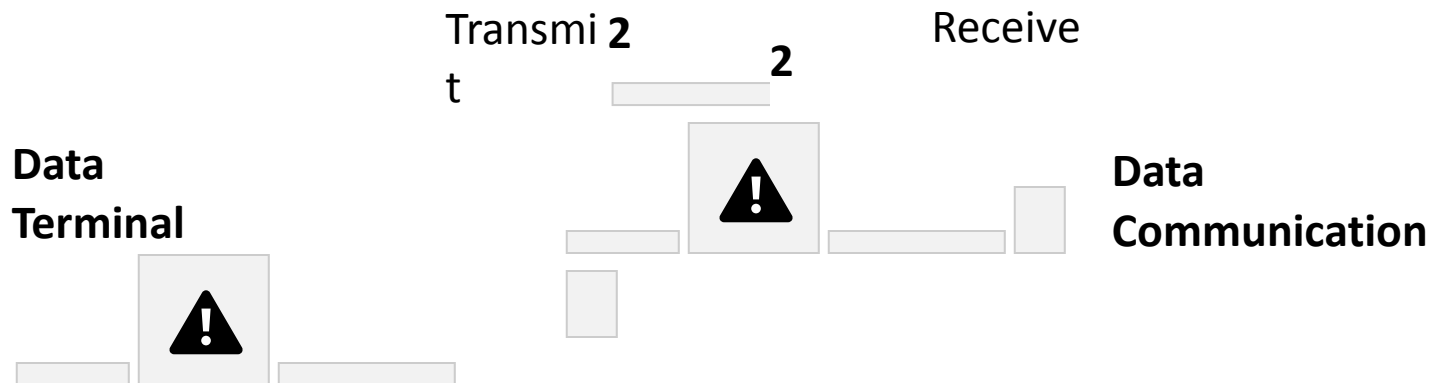
FE framing error (Aynsc only, Valid stop bit not detected)

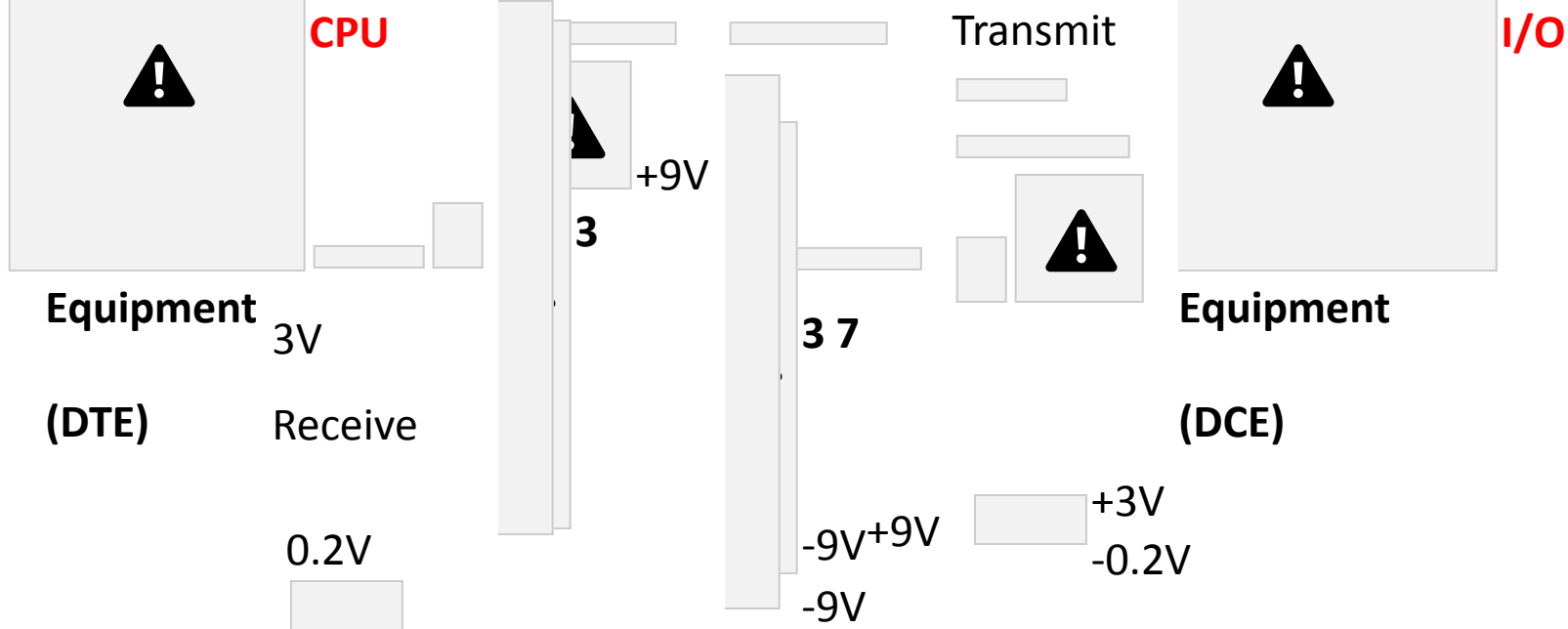
SYNDET sync. character detected

DSR data set ready (DSR set at 0 level)



RS-232 Cable





- RS232: Data transmitted as Voltage to terminal
 - 20KBps, 50Mters only
 - Improved to RS 422A (9 pine), RS 423A (15 pin-VGA)
- Modem (Data transmitted by Frequency)



2	TxD: transmitted Data	Output CPU to I/O
3	RxD :Received Data	Input I/O receive from
4	RTS :Request to Send	Output from I/O
5	CTS :Clear to send	Input to I/O, HS sign
6	DSR: Data set ready	CPU send to I/O is re
7	GND	Comm. Ref GND
8	DCD: Data Carrier Detect	I/O to disable recep
20	DTR: Data terminal ready	Output to indicate I/ is ready

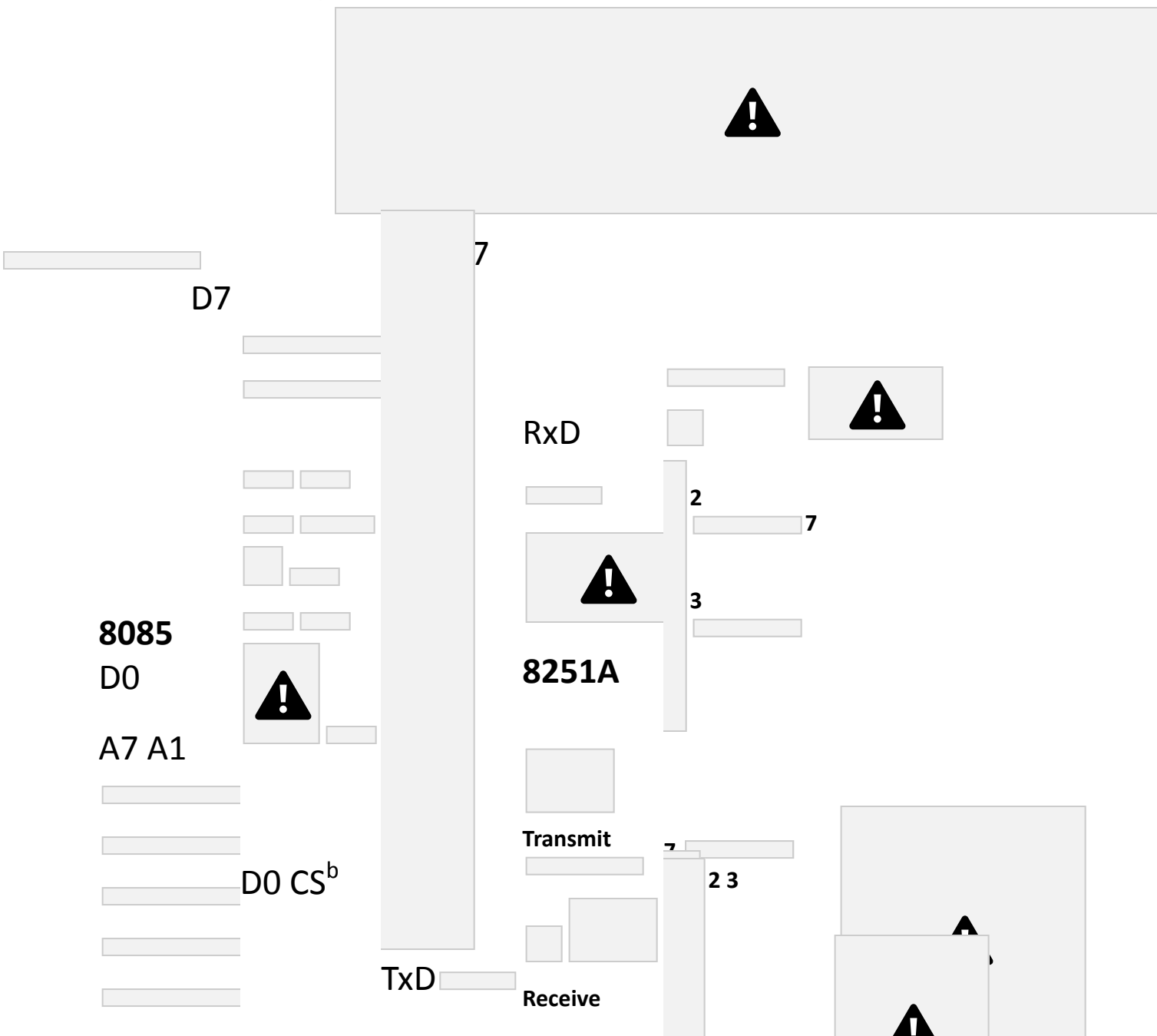


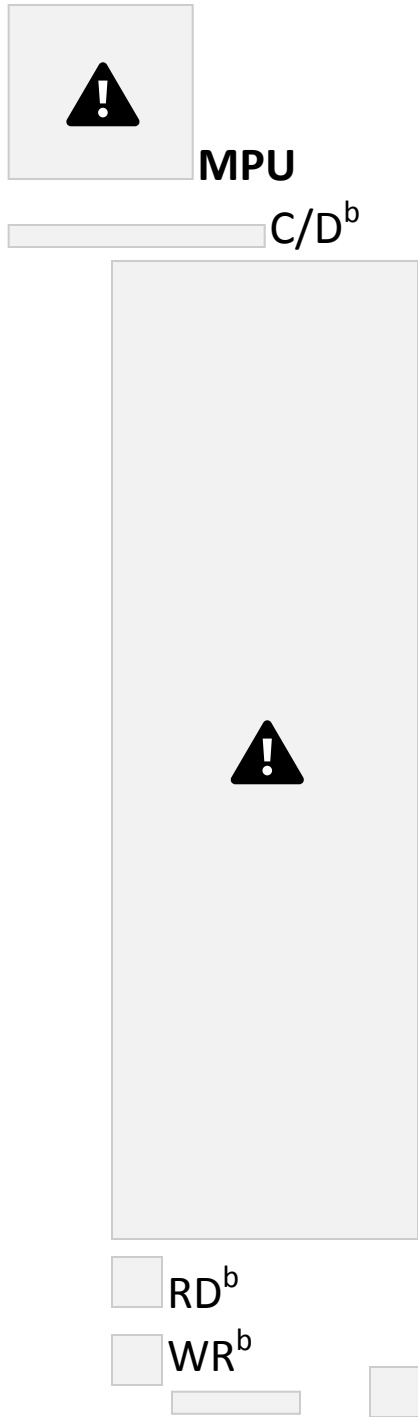
DB-25 DB9



- Connect a RS 232 port onto a CRT terminal • Address the 8251A USART at FF to control transmission
- Specify initialization instructions and status word to transmit characters
 - Async mode with 9600 buad
 - Character length= 7 bit + 2 stop bit
 - No parity check
- Write instruction to initialize USART and read status word and Setup a loop until the transmitter

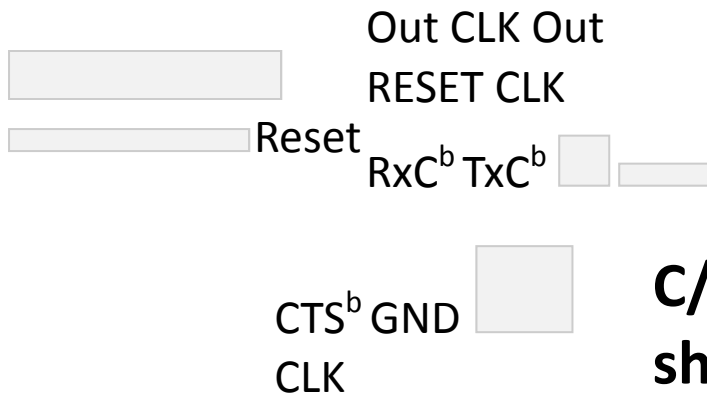
is ready





A0 Voltage Converter IOR^b





**Control &
Status**

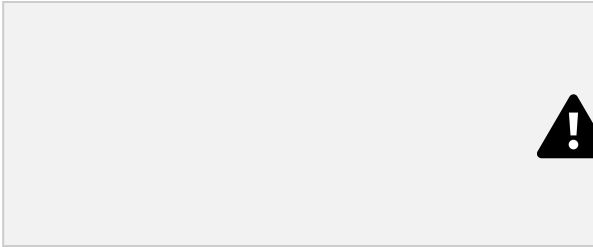
**Register
Address=FFH**

**C/D^bline
should be**

**high, == > A₀
=1**

CAH

Mode Word



COMMAND WORD

STATUS

D7	D6	D5
1	1	0
Two Stop bits		1

X	0

X	X	X

11H

01H



SETUP: MVI A,CAH ; *load mode word*

OUT FFH ; *Write mode word in control register* MVI A,11H
; *load command word to enable TX* OUT FFH ; *Enable the*
transmitter

STATUS: IN FFH ; *Read the status register* ANI 01H ;
Mask all bit except D0

JZ STATUS ; *if D0=0 the TX buffer is full*



Message is “HELLO CS421” 2070 0E ; 13 characters to follow

2071 48; Letter H

2072 45; Letter E

2073 4C; Letter L

2074 4C ; Letter L

2075 4F ; Letter H

2076 20; space

2077 43; Letter C

2078 53; Letter S

2079 34; Digit 4

2080 32; Digit 2

2081 31; Digit 1

2082 0D; Carriage return

2083 0A; Linefeed



LXI H 2070H ; Memory ptr for Message MOV C, M ; Set up Ctr register
MVI A,40; Reset 8251
OUT FFH

```
MVI A,CA; Initialize 82512  
OUT FFH  
MVI A,11 ; initialize for transmit  
OUT FFH
```



```
STATUS: IN FFH  
ANI 01H ;Ckeck TxRDY  
JZ STATUS ; is txRDY 1 ? If not wait  
INX H ; Pont to Next Char  
MOV A,M ; place the Char in ACC  
OUT FEH ;  
Send the  
Char to
```

```
Transmitter DCR C ; DCr cnt  
JNZ STATUS ;Again Send the rest of Char HLT
```